

Ex.no:4b	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

AIM

To perform data inspection, filtering, sorting, grouping, aggregation, descriptive statistical analysis, and correlation analysis on the Life Expectancy dataset using Pandas, and to visualize the analytical results using Matplotlib and Seaborn.

ALGORITHM

Algorithm 1: Dataset Inspection

1. Import the Life Expectancy dataset.
2. Display the first and last few records of the dataset.
3. Check the dimensions of the dataset.
4. Display the data types of the attributes.
5. Generate a statistical summary of the numerical attributes.

Algorithm 2: Filtering Data

1. Select the required attribute.
2. Specify a suitable filtering condition.
3. Apply the condition to the dataset.
4. Display the filtered records.

Algorithm 3: Sorting Data

1. Select the attribute to be sorted.
2. Apply the `sort_values()` function.
3. Specify the sorting order.
4. Display the sorted dataset.

Algorithm 4: Statistical Analysis

1. Select the required numerical attribute.
2. Calculate the mean.
3. Calculate the median.
4. Calculate the mode.
5. Calculate the standard deviation.
6. Display the calculated statistical measures.

Algorithm 5: Grouping and Aggregation

1. Select a categorical attribute such as Country or Status.
2. Apply the `groupby()` function.

3. Select the required numerical attribute.
4. Calculate an aggregate measure such as mean.
5. Display the grouped results.

Algorithm 6: Correlation Analysis

1. Select the required numerical attributes.
2. Apply the corr() function.
3. Generate the correlation matrix.
4. Analyze the correlation coefficient between the selected attributes.
5. Visualize the relationship using an appropriate graph.

CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os

# =====
# LOAD LIFE EXPECTANCY DATASET
# =====

path = "/content"

print("Dataset Path:")
print(path)

print("\nFiles Found:")
print(os.listdir(path))

# Find CSV file automatically
csv_files = [f for f in os.listdir(path) if f.lower().endswith(".csv")]

if not csv_files:
    raise FileNotFoundError("No CSV file found in the dataset path.")

csv_file = csv_files[0]

# Load Dataset
df = pd.read_csv(
    os.path.join(path, csv_file),
    encoding="latin1",
    low_memory=False
)

print("\nDataset Loaded Successfully")

# =====
# PROGRAM 1 - DATA INSPECTION
# =====

print("\nFirst 5 Records")
print(df.head())
```

```
print("\nLast 5 Records")
print(df.tail())
```

```
# =====
# PROGRAM 2 - SHAPE
# =====
```

```
print("\nShape of Dataset")
print(df.shape)
```

```
print("\nRows =", df.shape[0])
print("\nColumns =", df.shape[1])
```

```
# =====
# PROGRAM 3 - DATA TYPES
# =====
```

```
print("\nData Types")
print(df.dtypes)
```

```
print("\nDataset Info")
df.info()
```

```
# =====
# PROGRAM 4 - FILTERING
# =====
```

```
# Countries with Life Expectancy greater than 75
```

```
high_life = df[df["Life expectancy"] > 75]
```

```
print("\nCountries with Life Expectancy Greater Than 75")
```

```
print(
    high_life[
        [
            "Country",
            "Year",
            "Status",
            "Life expectancy",
            "GDP",
            "Schooling"
        ]
    ].head(10)
)
```

```
# =====
# PROGRAM 5 - FILTER LOW LIFE EXPECTANCY
# =====
```

```
low_life = df[df["Life expectancy"] < 60]
```

```
print("\nRecords with Life Expectancy Less Than 60")
```

```
print(
    low_life[
        [
            "Country",
            "Year",
            "Status",
            "Life expectancy",
            "GDP",
            "Schooling"
        ]
    ].head(10)
)
```

```
# =====
# PROGRAM 6 - SORTING
# =====
```

```
# Sort by Life Expectancy
```

```
sorted_df = df.sort_values(
    by="Life expectancy",
    ascending=False
)
```

```
print("\nTop Countries by Life Expectancy")
```

```
print(
    sorted_df[
        [
            "Country",
            "Year",
            "Status",
            "Life expectancy",
            "GDP",
            "Schooling"
        ]
    ].head(10)
)
```

```
# =====
# BAR CHART
# TOP 10 LIFE EXPECTANCY RECORDS
# =====
```

```
top10 = sorted_df.head(10)
```

```
plt.figure(figsize=(10, 5))
```

```
plt.bar(
    range(len(top10)),
    top10["Life expectancy"]
)
```

```

plt.xticks(
    range(len(top10)),
    top10["Country"],
    rotation=90
)

plt.xlabel("Country")
plt.ylabel("Life Expectancy")
plt.title("Top 10 Records by Life Expectancy")

plt.show()

```

```

# =====
# PROGRAM 7 - STATISTICAL ANALYSIS
# =====

```

```

print("\nMean Life Expectancy =")
print(df["Life expectancy"].mean())

print("\nMedian Life Expectancy =")
print(df["Life expectancy"].median())

print("\nMode Life Expectancy =")
print(df["Life expectancy"].mode())

print("\nStandard Deviation =")
print(df["Life expectancy"].std())

print("\nMinimum Life Expectancy =")
print(df["Life expectancy"].min())

print("\nMaximum Life Expectancy =")
print(df["Life expectancy"].max())

```

```

# =====
# PROGRAM 8 - DESCRIPTIVE STATISTICS
# =====

```

```

print("\nDescriptive Statistics")

```

```

print(
    df[
        "Life expectancy",
        "Adult Mortality",
        "infant deaths",
        "GDP",
        "Population",
        "Schooling"
    ]
    .describe()
)

```

```

# =====

```

```
# PROGRAM 9 - GROUPING
```

```
# =====
```

```
# Average Life Expectancy by Status
```

```
group = df.groupby(  
    "Status"  
)[ "Life expectancy"].mean().sort_values(ascending=False)
```

```
print("\nAverage Life Expectancy by Status")  
print(group)
```

```
# Visualization
```

```
group.plot(  
    kind="bar",  
    figsize=(8, 5)  
)
```

```
plt.xlabel("Status")  
plt.ylabel("Average Life Expectancy")  
plt.title("Average Life Expectancy by Status")  
plt.xticks(rotation=0)
```

```
plt.show()
```

```
# =====
```

```
# PROGRAM 10 - AGGREGATION
```

```
# =====
```

```
summary = df.groupby("Status").agg({
```

```
    "Life expectancy": "mean",  
    "Adult Mortality": "mean",  
    "GDP": "mean",  
    "Schooling": "mean"
```

```
})
```

```
print("\nStatus Aggregation")  
print(summary)
```

```
# =====
```

```
# PROGRAM 11 - CORRELATION
```

```
# =====
```

```
numeric = df.select_dtypes(  
    include=np.number  
)
```

```
corr = numeric.corr()
```

```
print("\nCorrelation Matrix")  
print(corr)
```

```
plt.figure(figsize=(12, 8))

sns.heatmap(
    corr,
    annot=False,
    cmap="coolwarm"
)

plt.title("Life Expectancy Dataset Correlation Heatmap")

plt.show()
```

```
# =====
# PROGRAM 12 - SCATTER PLOT
# Adult Mortality vs Life Expectancy
# =====
```

```
plt.figure(figsize=(7, 5))

plt.scatter(
    df["Adult Mortality"],
    df["Life expectancy"]
)

plt.xlabel("Adult Mortality")
plt.ylabel("Life Expectancy")

plt.title("Adult Mortality vs Life Expectancy")

plt.show()
```

```
# =====
# PROGRAM 13 - HISTOGRAM
# Life Expectancy Distribution
# =====
```

```
plt.figure(figsize=(7, 5))

plt.hist(
    df["Life expectancy"].dropna(),
    bins=30
)

plt.xlabel("Life Expectancy")
plt.ylabel("Frequency")

plt.title("Distribution of Life Expectancy")

plt.show()
```

```
# =====
# PROGRAM 14 - BOXPLOT
# =====
```

```
plt.figure(figsize=(7, 5))

sns.boxplot(
    x=df["Life expectancy"].dropna()
)

plt.title("Life Expectancy Boxplot")
plt.xlabel("Life Expectancy")

plt.show()
```

```
# =====
# PROGRAM 15 - VALUE COUNTS
# =====
```

```
print("\nStatus Counts")

print(
    df["Status"].value_counts()
)
```

```
# =====
# PROGRAM 16 - UNIQUE VALUES
# =====
```

```
print("\nUnique Countries")

print(
    df["Country"].unique()
)

print("\nUnique Status Values")

print(
    df["Status"].unique()
)

print("\nUnique Years")

print(
    df["Year"].unique()
)
```

```
# =====
# PROGRAM 17 - TOP 10 COUNTRIES BY LIFE EXPECTANCY
# =====
```

```
print("\nTop 10 Countries by Average Life Expectancy")

country_life = (
    df.groupby("Country")["Life expectancy"]
    .mean()
    .sort_values(ascending=False)
```

```
.head(10)
)

print(country_life)
```

```
# =====
# PROGRAM 18 - LIFE EXPECTANCY BY YEAR
# =====
```

```
print("\nAverage Life Expectancy by Year")
```

```
year_life = (
    df.groupby("Year")["Life expectancy"]
    .mean()
    .sort_index()
)

print(year_life)
```

```
# =====
# PROGRAM 19 - TOP 10 COUNTRIES BY LOW LIFE EXPECTANCY
# =====
```

```
print("\nBottom 10 Countries by Average Life Expectancy")
```

```
country_low = (
    df.groupby("Country")["Life expectancy"]
    .mean()
    .sort_values()
    .head(10)
)

print(country_low)
```

```
# =====
# PROGRAM 20 - MOST COMMON STATUS
# =====
```

```
print("\nStatus Distribution")
```

```
status_counts = (
    df["Status"]
    .value_counts()
    .head(10)
)

print(status_counts)
```

```
# =====
# PROGRAM 21 - EXPORT DATA
# =====
```

```
df.to_csv(
```

```

    "life_expectancy_data.csv",
    index=False
)

df.to_excel(
    "life_expectancy_data.xlsx",
    index=False
)

df.to_json(
    "life_expectancy_data.json",
    orient="records",
    indent=4
)

print("\nCSV, Excel and JSON files exported successfully.")

# =====
# FINAL REPORT
# =====

print("\n=====")
print("EXPERIMENT COMPLETED SUCCESSFULLY")
print("=====")

print("Dataset Shape :", df.shape)

print(
    "Missing Values :",
    df.isnull().sum().sum()
)

print(
    "Total Records :",
    len(df)
)

print(
    "Average Life Expectancy :",
    df["Life expectancy"].mean()
)

print(
    "Number of Countries :",
    df["Country"].nunique()
)

print(
    "Number of Years :",
    df["Year"].nunique()
)

print(
    "Number of Status Types :",
    df["Status"].nunique()
)

```

```
print("Correlation Matrix Generated")
```

```
print("=====")
```

OUTPUT:

```
Country Year Status Life expectancy Adult Mortality \
```

```
0 Afghanistan 2015 Developing 65.0 263.0
1 Afghanistan 2014 Developing 59.9 271.0
2 Afghanistan 2013 Developing 59.9 268.0
3 Afghanistan 2012 Developing 59.5 272.0
4 Afghanistan 2011 Developing 59.2 275.0
```

```
infant deaths Alcohol percentage expenditure Hepatitis B Measles ... \
0 62 0.01 71.279624 65.0 1154 ...
1 64 0.01 73.523582 62.0 492 ...
2 66 0.01 73.219243 64.0 430 ...
3 69 0.01 78.184215 67.0 2787 ...
4 71 0.01 7.097109 68.0 3013 ...
```

```
Polio Total expenditure Diphtheria HIV/AIDS GDP Population \
0 6.0 8.16 65.0 0.1 584.259210 33736494.0
1 58.0 8.18 62.0 0.1 612.696514 327582.0
2 62.0 8.13 64.0 0.1 631.744976 31731688.0
3 67.0 8.52 67.0 0.1 669.959000 3696958.0
4 68.0 7.87 68.0 0.1 63.537231 2978599.0
```

```
thinness 1-19 years thinness 5-9 years Income composition of resources \
0 17.2 17.3 0.479
1 17.5 17.5 0.476
2 17.7 17.7 0.470
3 17.9 18.0 0.463
4 18.2 18.2 0.454
```

```
Schooling
0 10.1
1 10.0
2 9.9
3 9.8
4 9.5
```

```
[5 rows x 22 columns]
```

```
Dataset Shape : (2938, 22)
```

```
Dataset Information:
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 2938 entries, 0 to 2937
```

```
Data columns (total 22 columns):
```

```
# Column Non-Null Count Dtype
---
0 Country 2938 non-null object
1 Year 2938 non-null int64
2 Status 2938 non-null object
3 Life expectancy 2928 non-null float64
4 Adult Mortality 2928 non-null float64
5 infant deaths 2938 non-null int64
6 Alcohol 2744 non-null float64
```

```

7 percentage expenditure      2938 non-null float64
8 Hepatitis B                 2385 non-null float64
9 Measles                     2938 non-null int64
10 BMI                        2904 non-null float64
11 under-five deaths          2938 non-null int64
12 Polio                      2919 non-null float64
13 Total expenditure          2712 non-null float64
14 Diphtheria                 2919 non-null float64
15 HIV/AIDS                   2938 non-null float64
16 GDP                        2490 non-null float64
17 Population                 2286 non-null float64
18 thinness 1-19 years        2904 non-null float64
19 thinness 5-9 years         2904 non-null float64
20 Income composition of resources 2771 non-null float64
21 Schooling                  2775 non-null float64
dtypes: float64(16), int64(4), object(2)
memory usage: 505.1+ KB

```

Data Types:

```

Country      object
Year         int64
Status       object
Life expectancy float64
Adult Mortality float64
infant deaths int64
Alcohol      float64
percentage expenditure float64
Hepatitis B  float64
Measles      int64
BMI          float64
under-five deaths int64
Polio        float64
Total expenditure float64
Diphtheria   float64
HIV/AIDS     float64
GDP          float64
Population   float64
thinness 1-19 years float64
thinness 5-9 years float64
Income composition of resources float64
Schooling    float64
dtype: object

```

Statistical Summary:

	Year	Life expectancy	Adult Mortality	infant deaths \
count	2938.000000	2928.000000	2928.000000	2938.000000
mean	2007.518720	69.224932	164.796448	30.303948
std	4.613841	9.523867	124.292079	117.926501
min	2000.000000	36.300000	1.000000	0.000000
25%	2004.000000	63.100000	74.000000	0.000000
50%	2008.000000	72.100000	144.000000	3.000000
75%	2012.000000	75.700000	228.000000	22.000000
max	2015.000000	89.000000	723.000000	1800.000000

	Alcohol	percentage expenditure	Hepatitis B	Measles \
count	2744.000000	2938.000000	2385.000000	2938.000000
mean	4.602861	738.251295	80.940461	2419.592240
std	4.052413	1987.914858	25.070016	11467.272489
min	0.010000	0.000000	1.000000	0.000000
25%	0.877500	4.685343	77.000000	0.000000

50%	3.755000	64.912906	92.000000	17.000000
75%	7.702500	441.534144	97.000000	360.250000
max	17.870000	19479.911610	99.000000	212183.000000

	BMI	under-five deaths	Polio	Total expenditure \
count	2904.000000	2938.000000	2919.000000	2712.000000
mean	38.321247	42.035739	82.550188	5.93819
std	20.044034	160.445548	23.428046	2.49832
min	1.000000	0.000000	3.000000	0.37000
25%	19.300000	0.000000	78.000000	4.26000
50%	43.500000	4.000000	93.000000	5.75500
75%	56.200000	28.000000	97.000000	7.49250
max	87.300000	2500.000000	99.000000	17.60000

	Diphtheria	HIV/AIDS	GDP	Population \
count	2919.000000	2938.000000	2490.000000	2.286000e+03
mean	82.324084	1.742103	7483.158469	1.275338e+07
std	23.716912	5.077785	14270.169342	6.101210e+07
min	2.000000	0.100000	1.681350	3.400000e+01
25%	78.000000	0.100000	463.935626	1.957932e+05
50%	93.000000	0.100000	1766.947595	1.386542e+06
75%	97.000000	0.800000	5910.806335	7.420359e+06
max	99.000000	50.600000	119172.741800	1.293859e+09

	thinness 1-19 years	thinness 5-9 years \
count	2904.000000	2904.000000
mean	4.839704	4.870317
std	4.420195	4.508882
min	0.100000	0.100000
25%	1.600000	1.500000
50%	3.300000	3.300000
75%	7.200000	7.200000
max	27.700000	28.600000

	Income composition of resources	Schooling
count	2771.000000	2775.000000
mean	0.627551	11.992793
std	0.210904	3.358920
min	0.000000	0.000000
25%	0.493000	10.100000
50%	0.677000	12.300000
75%	0.779000	14.300000
max	0.948000	20.700000

Missing Values:

Country	0
Year	0
Status	0
Life expectancy	10
Adult Mortality	10
infant deaths	0
Alcohol	194
percentage expenditure	0
Hepatitis B	553
Measles	0
BMI	34
under-five deaths	0
Polio	19
Total expenditure	226
Diphtheria	19

HIV/AIDS	0
GDP	448
Population	652
thinness 1-19 years	34
thinness 5-9 years	34
Income composition of resources	167
Schooling	163

dtype: int64

Missing Value Percentage:

Country	0.000000
Year	0.000000
Status	0.000000
Life expectancy	0.340368
Adult Mortality	0.340368
infant deaths	0.000000
Alcohol	6.603131
percentage expenditure	0.000000
Hepatitis B	18.822328
Measles	0.000000
BMI	1.157250
under-five deaths	0.000000
Polio	0.646698
Total expenditure	7.692308
Diphtheria	0.646698
HIV/AIDS	0.000000
GDP	15.248468
Population	22.191967
thinness 1-19 years	1.157250
thinness 5-9 years	1.157250
Income composition of resources	5.684139
Schooling	5.547992

dtype: float64

Duplicate Rows : 0

After Duplicate Removal : (2938, 22)

Missing Values After Imputation:

Country	0
Year	0
Status	0
Life expectancy	0
Adult Mortality	0
infant deaths	0
Alcohol	0
percentage expenditure	0
Hepatitis B	0
Measles	0
BMI	0
under-five deaths	0
Polio	0
Total expenditure	0
Diphtheria	0
HIV/AIDS	0
GDP	0
Population	0
thinness 1-19 years	0
thinness 5-9 years	0
Income composition of resources	0
Schooling	0

dtype: int64

Status Categories:

['Developing' 'Developed']

Number of Countries:

193

Years:

[2015 2014 2013 2012 2011 2010 2009 2008 2007 2006 2005 2004 2003 2002
2001 2000]

High Life Expectancy Records:

	Country	Year	Status	Life expectancy	Adult Mortality \
16	Albania	2015	Developing	77.8	74.0
17	Albania	2014	Developing	77.5	8.0
18	Albania	2013	Developing	77.2	84.0
19	Albania	2012	Developing	76.9	86.0
20	Albania	2011	Developing	76.6	88.0

	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles ... \
16	0	4.60	364.975229	99.0	0 ...
17	0	4.51	428.749067	98.0	0 ...
18	0	4.76	430.876979	99.0	0 ...
19	0	5.14	412.443356	99.0	9 ...
20	0	5.37	437.062100	99.0	28 ...

	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population \
16	99.0	6.00	99.0	0.1	3954.227830	28873.0
17	98.0	5.88	98.0	0.1	4575.763787	288914.0
18	99.0	5.66	99.0	0.1	4414.723140	289592.0
19	99.0	5.59	99.0	0.1	4247.614380	2941.0
20	99.0	5.71	99.0	0.1	4437.178680	295195.0

	thinness 1-19 years	thinness 5-9 years	Income composition of resources \
16	1.2	1.3	0.762
17	1.2	1.3	0.761
18	1.3	1.4	0.759
19	1.3	1.4	0.752
20	1.4	1.5	0.738

Schooling

16	14.2
17	14.2
18	14.2
19	14.2
20	13.3

[5 rows x 22 columns]

Low Life Expectancy Records:

	Country	Year	Status	Life expectancy	Adult Mortality \
1	Afghanistan	2014	Developing	59.9	271.0
2	Afghanistan	2013	Developing	59.9	268.0
3	Afghanistan	2012	Developing	59.5	272.0
4	Afghanistan	2011	Developing	59.2	275.0
5	Afghanistan	2010	Developing	58.8	279.0

	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles ... \
1	64	0.01	73.523582	62.0	492 ...

2	66	0.01	73.219243	64.0	430 ...
3	69	0.01	78.184215	67.0	2787 ...
4	71	0.01	7.097109	68.0	3013 ...
5	74	0.01	79.679367	66.0	1989 ...

	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population \
1	58.0	8.18	62.0	0.1	612.696514	327582.0
2	62.0	8.13	64.0	0.1	631.744976	31731688.0
3	67.0	8.52	67.0	0.1	669.959000	3696958.0
4	68.0	7.87	68.0	0.1	63.537231	2978599.0
5	66.0	9.20	66.0	0.1	553.328940	2883167.0

	thinness 1-19 years	thinness 5-9 years	Income composition of resources \
1	17.5	17.5	0.476
2	17.7	17.7	0.470
3	17.9	18.0	0.463
4	18.2	18.2	0.454
5	18.4	18.4	0.448

	Schooling
1	10.0
2	9.9
3	9.8
4	9.5
5	9.2

[5 rows x 22 columns]

Top Life Expectancy Records:

	Country	Year	Status	Life expectancy	Adult Mortality \
2056	Portugal	2014	Developed	89.0	78.0
995	Germany	2014	Developed	89.0	69.0
2433	Spain	2007	Developed	89.0	72.0
1850	New Zealand	2010	Developed	89.0	71.0
915	Finland	2014	Developing	89.0	78.0

	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles \
2056	0	9.88	271.254553	98.0	0
995	2	11.03	941.756291	88.0	443
2433	2	11.05	510.932701	96.0	267
1850	0	9.62	6664.148708	9.0	43
915	0	8.80	6164.455402	92.0	0

	...	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP \
2056	...	98.0	9.50	98.0	0.1	2277.53613
995	...	94.0	11.30	95.0	0.1	4792.65288
2433	...	96.0	8.36	96.0	0.1	3279.41400
1850	...	93.0	11.20	93.0	0.1	33691.34837
915	...	98.0	9.68	98.0	0.1	49914.61864

	Population	thinness 1-19 years	thinness 5-9 years \
2056	14162.0	0.7	0.5
995	89825.0	1.1	1.1
2433	4522683.0	0.6	0.5
1850	1386542.0	0.3	0.3
915	5461512.0	0.9	0.8

	Income composition of resources	Schooling
2056	0.837	16.8
995	0.920	17.0

2433	0.849	16.0
1850	0.899	20.3
915	0.890	17.0

[5 rows x 22 columns]

Mean Life Expectancy : 69.23471749489448

Median Life Expectancy : 72.1

Mode Life Expectancy:

0 73.0

Name: Life expectancy, dtype: float64

Standard Deviation : 9.509115008147452

Average Life Expectancy by Status:

Status

Developed 79.197852

Developing 67.132028

Name: Life expectancy, dtype: float64

Average Life Expectancy by Year:

Year

2000 66.750273

2001 67.128962

2002 67.351366

2003 67.433333

2004 67.646448

2005 68.209290

2006 68.667760

2007 69.036066

2008 69.427869

2009 69.938251

2010 70.048634

2011 70.654098

2012 70.916940

2013 71.280829

2014 71.536612

2015 71.616940

Name: Life expectancy, dtype: float64

Top 10 Countries:

Country

Japan 82.53750

Sweden 82.51875

Iceland 82.44375

Switzerland 82.33125

France 82.21875

Italy 82.18750

Spain 82.06875

Australia 81.81250

Norway 81.79375

Canada 81.68750

Name: Life expectancy, dtype: float64

Bottom 10 Countries:

Country

Sierra Leone 46.11250

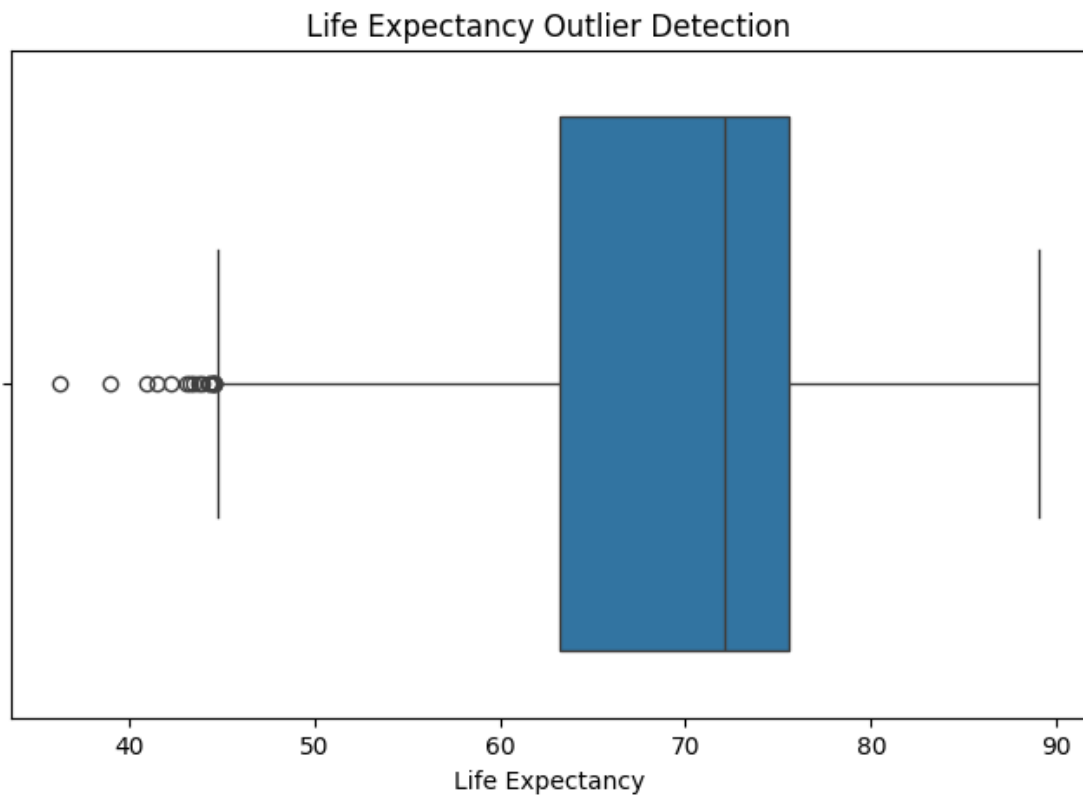
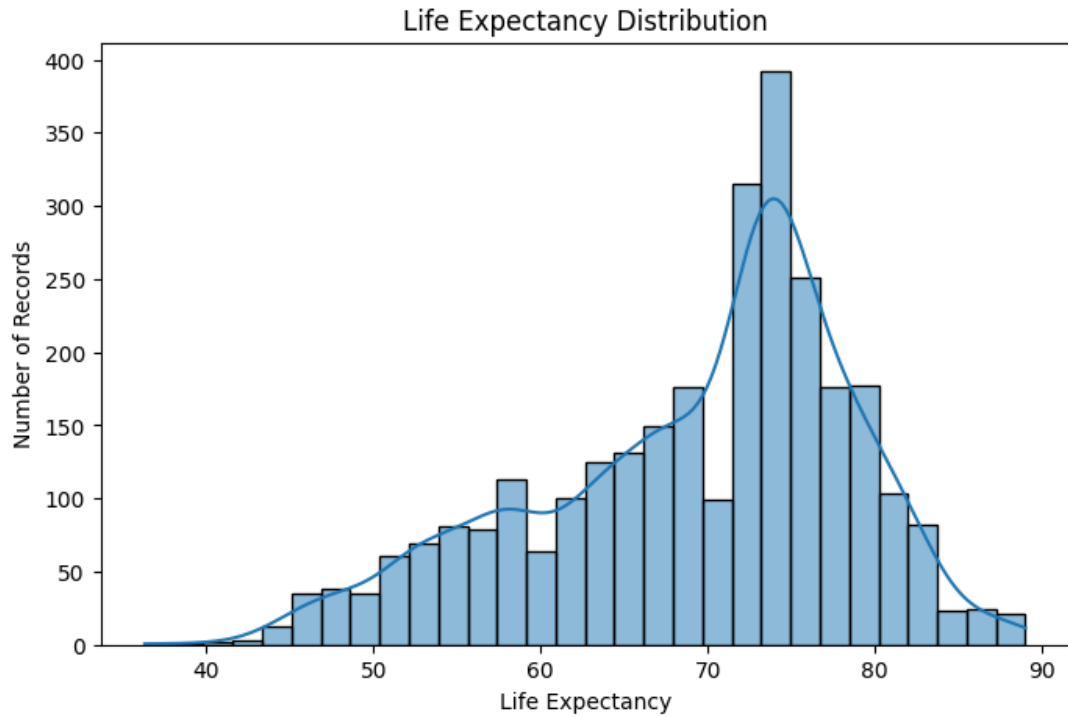
Central African Republic 48.51250

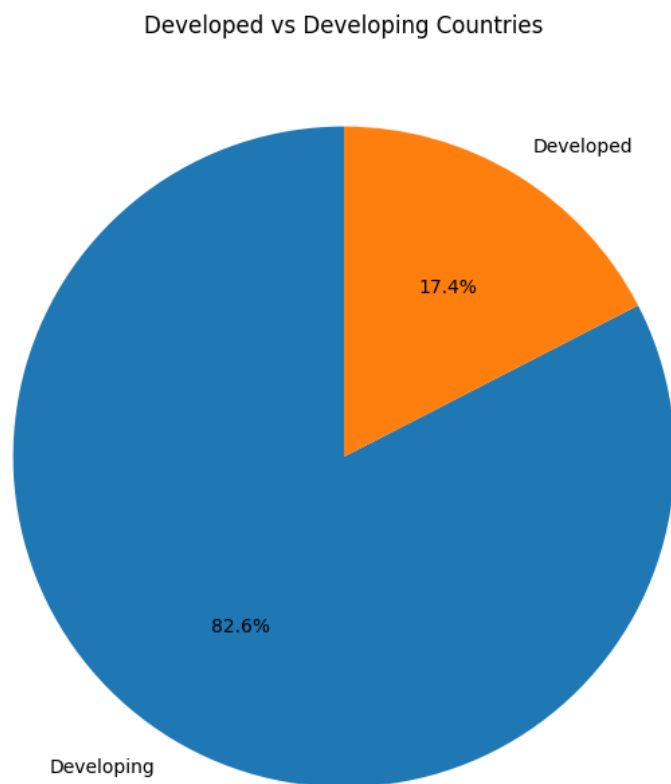
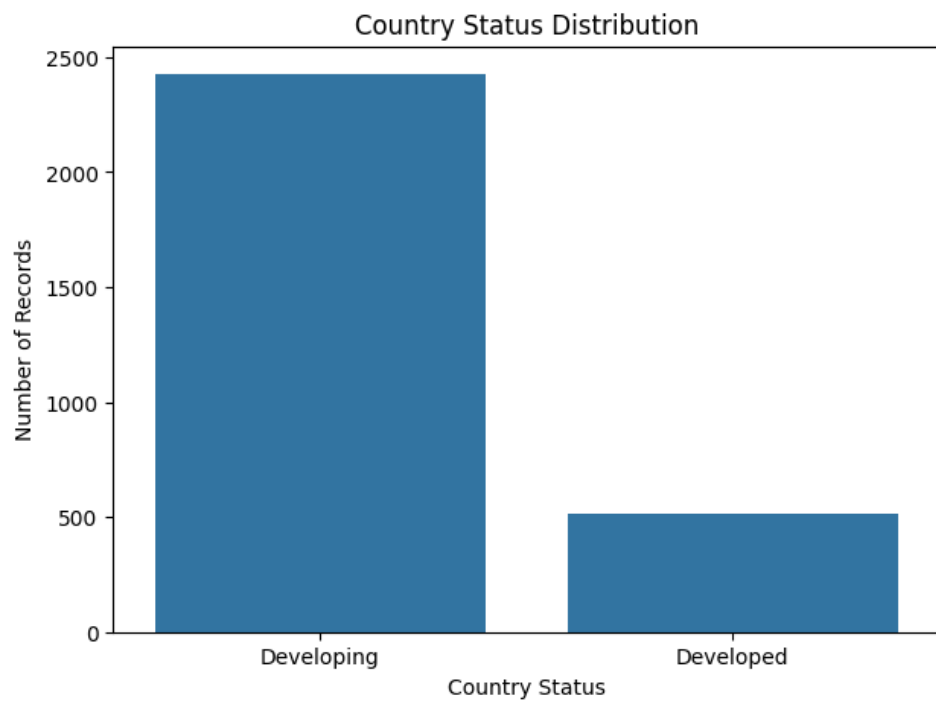
Lesotho 48.78125

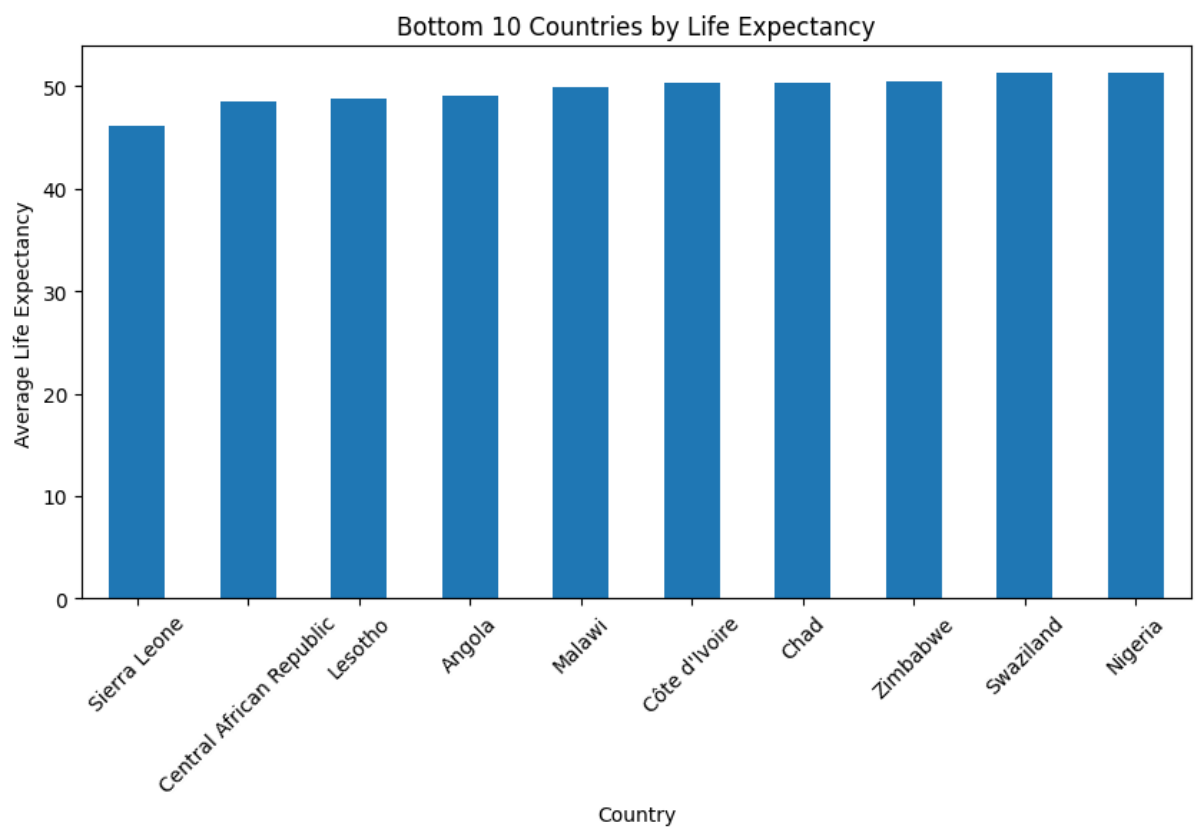
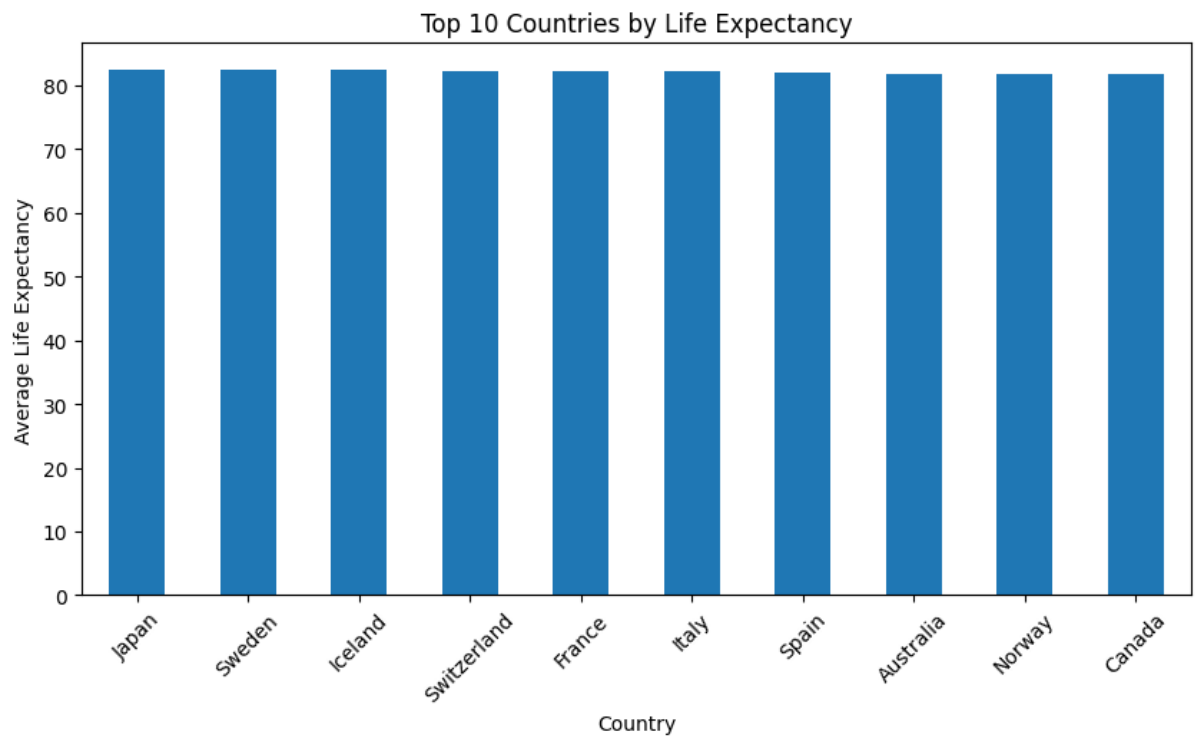
Angola 49.01875

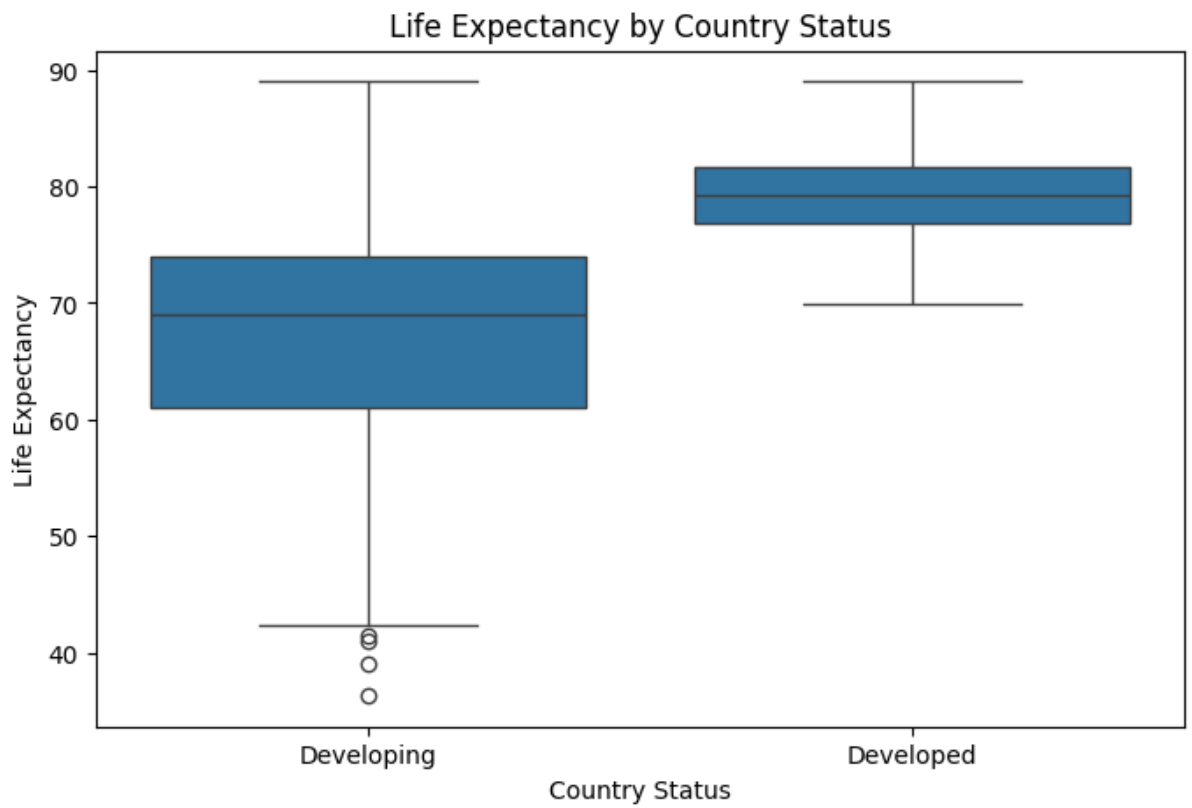
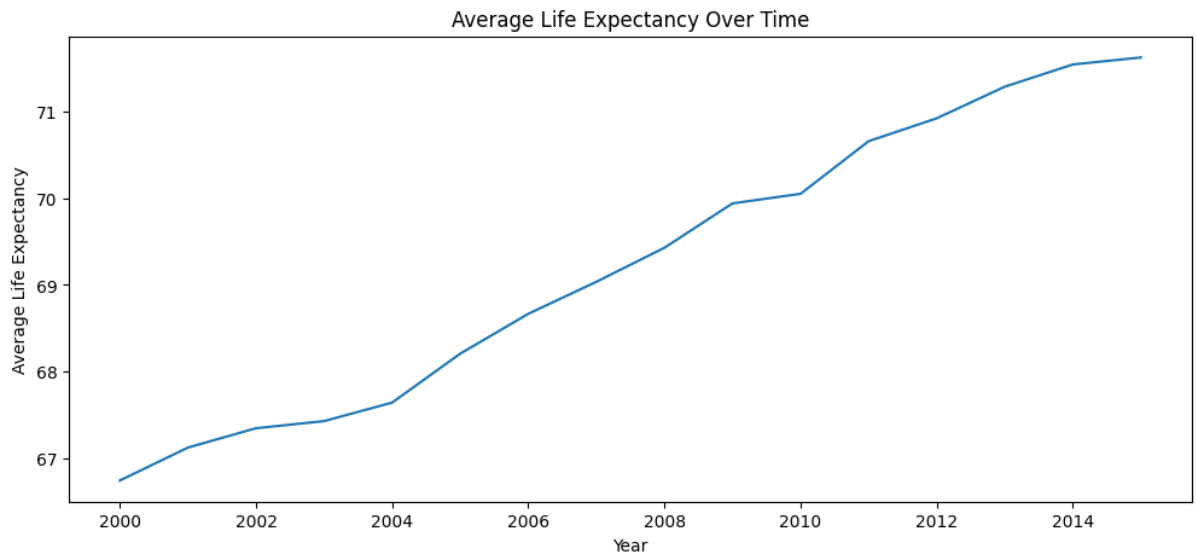
Malawi	49.89375
Côte d'Ivoire	50.38750
Chad	50.38750
Zimbabwe	50.48750
Swaziland	51.32500
Nigeria	51.35625

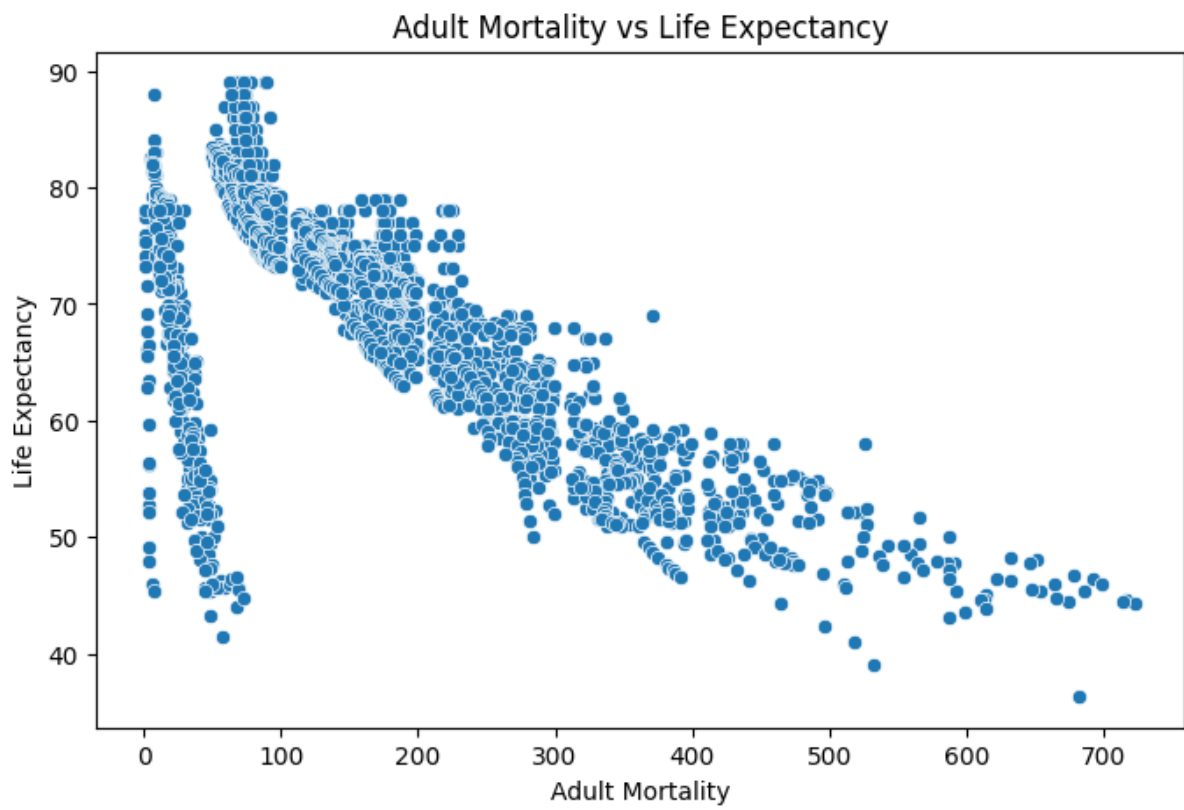
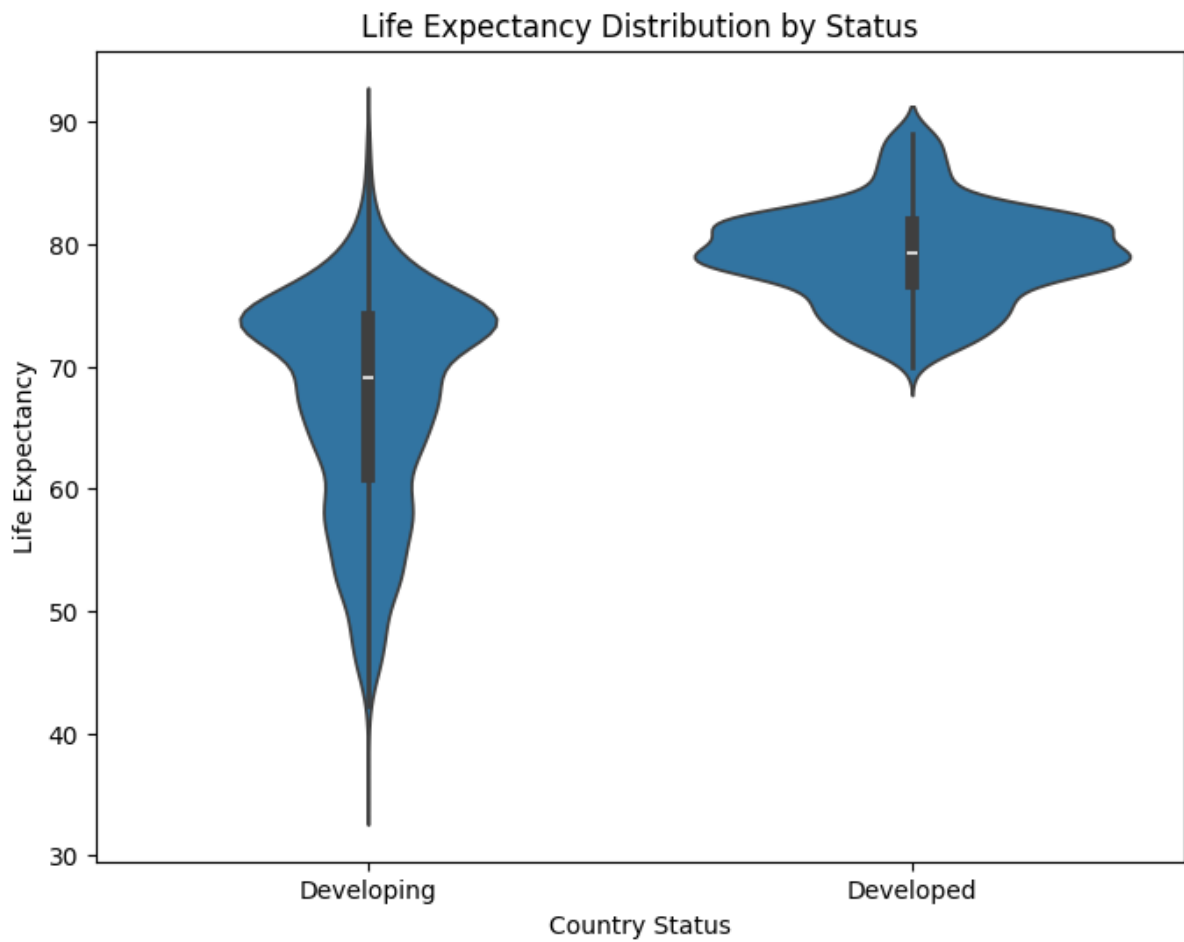
Name: Life expectancy, dtype: float64

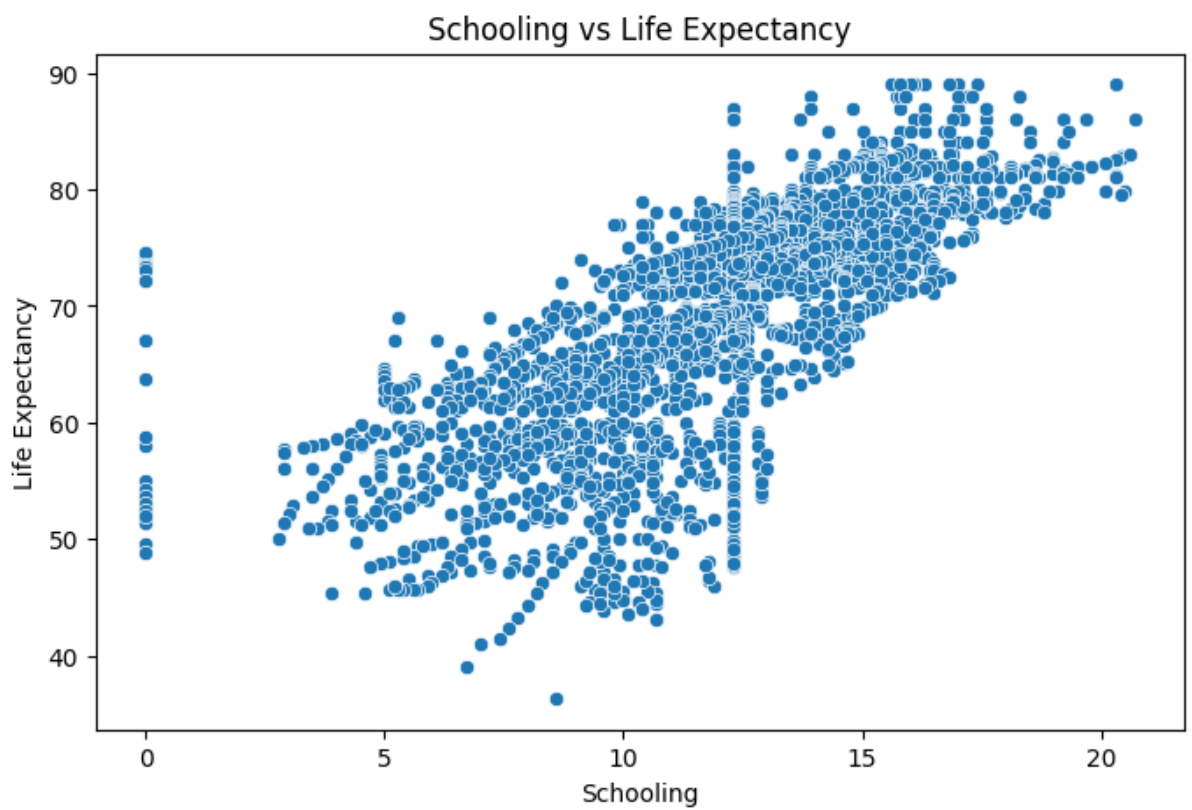
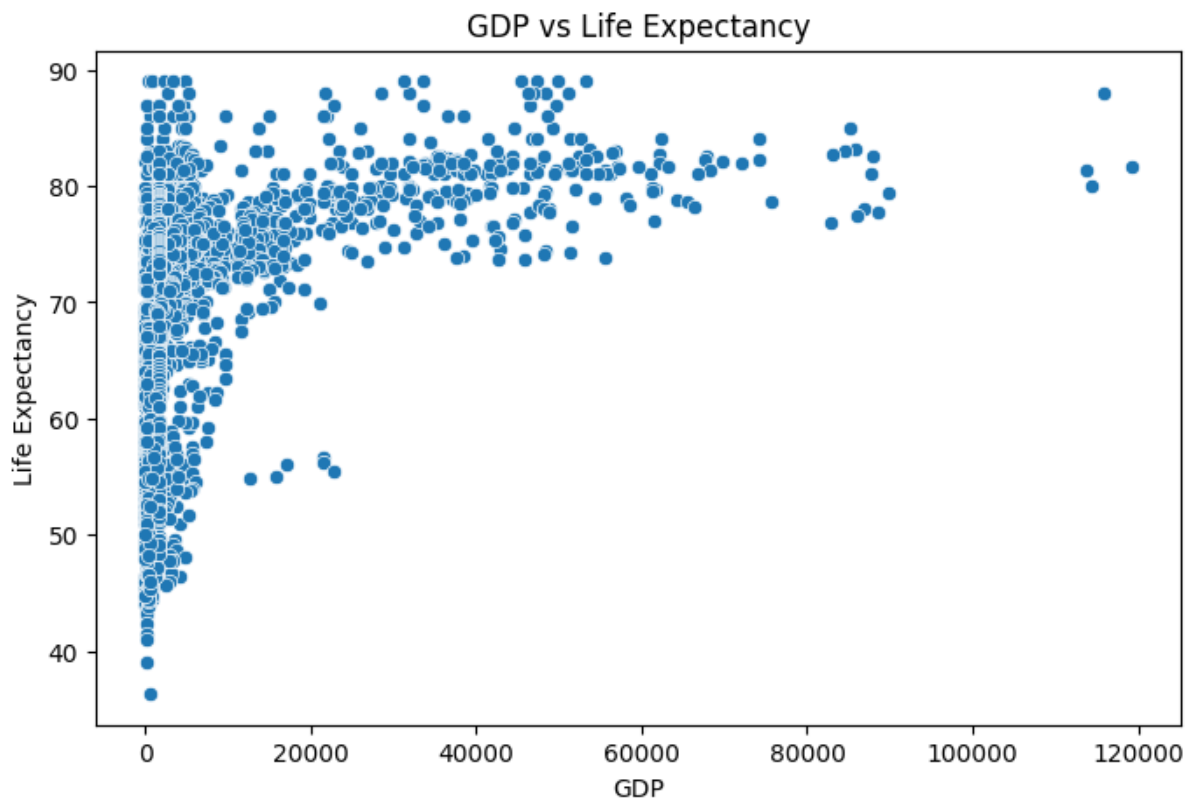


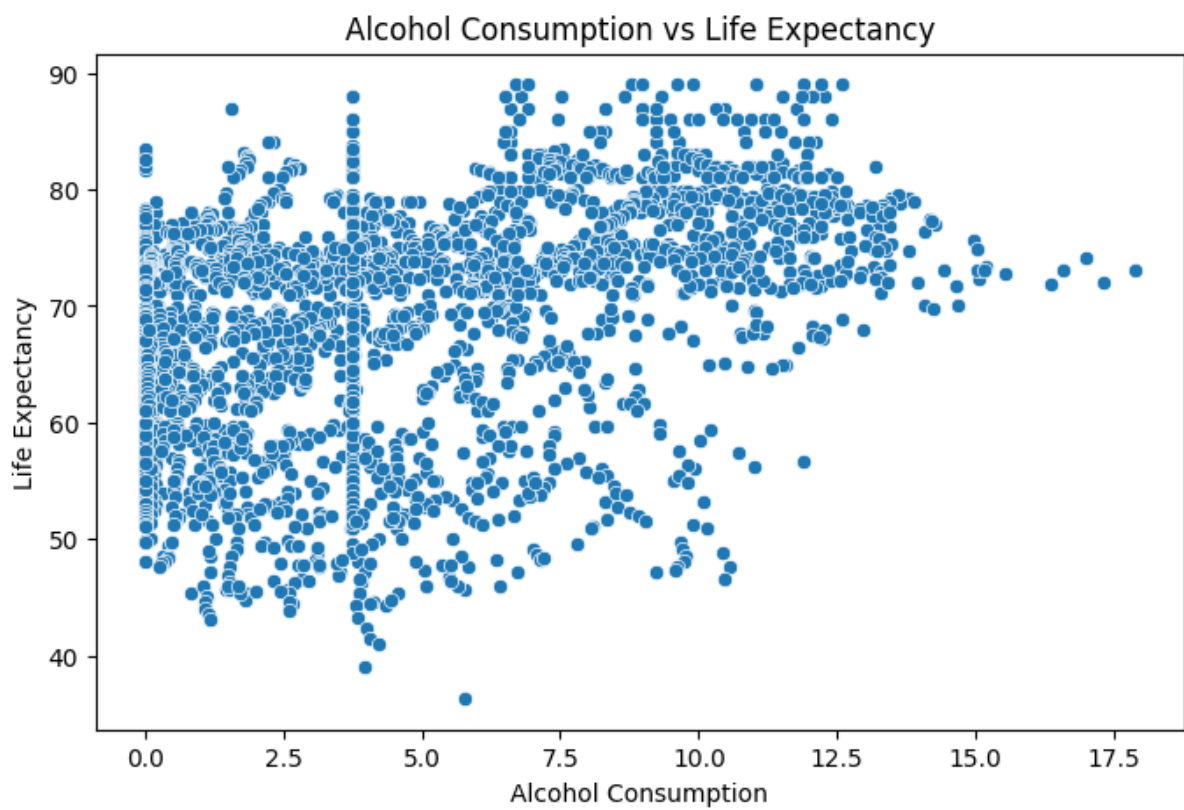
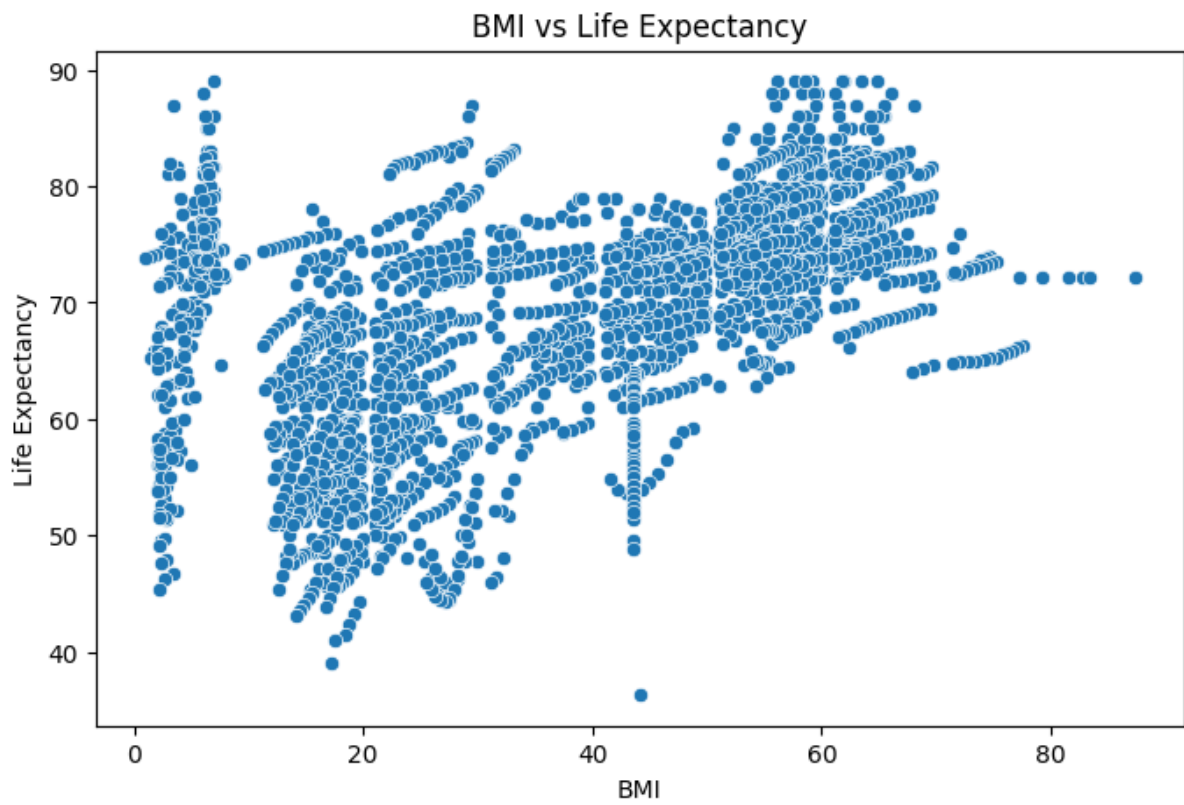


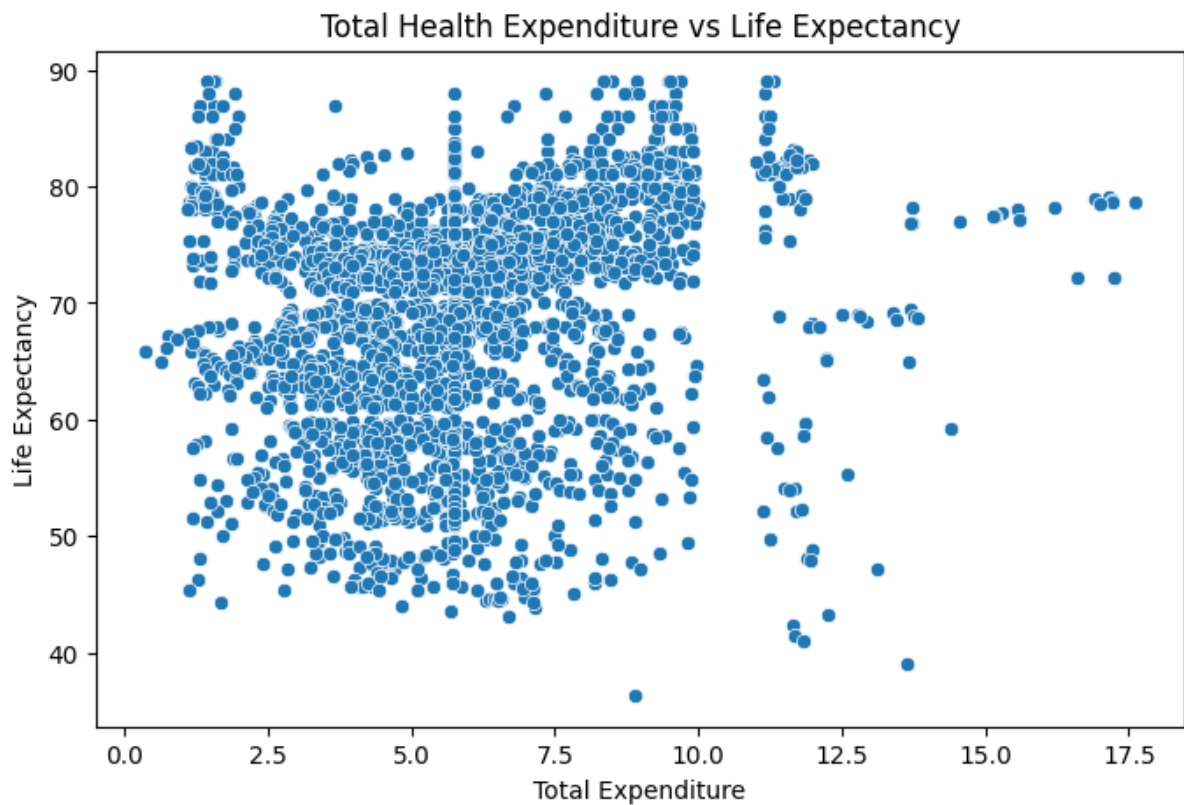
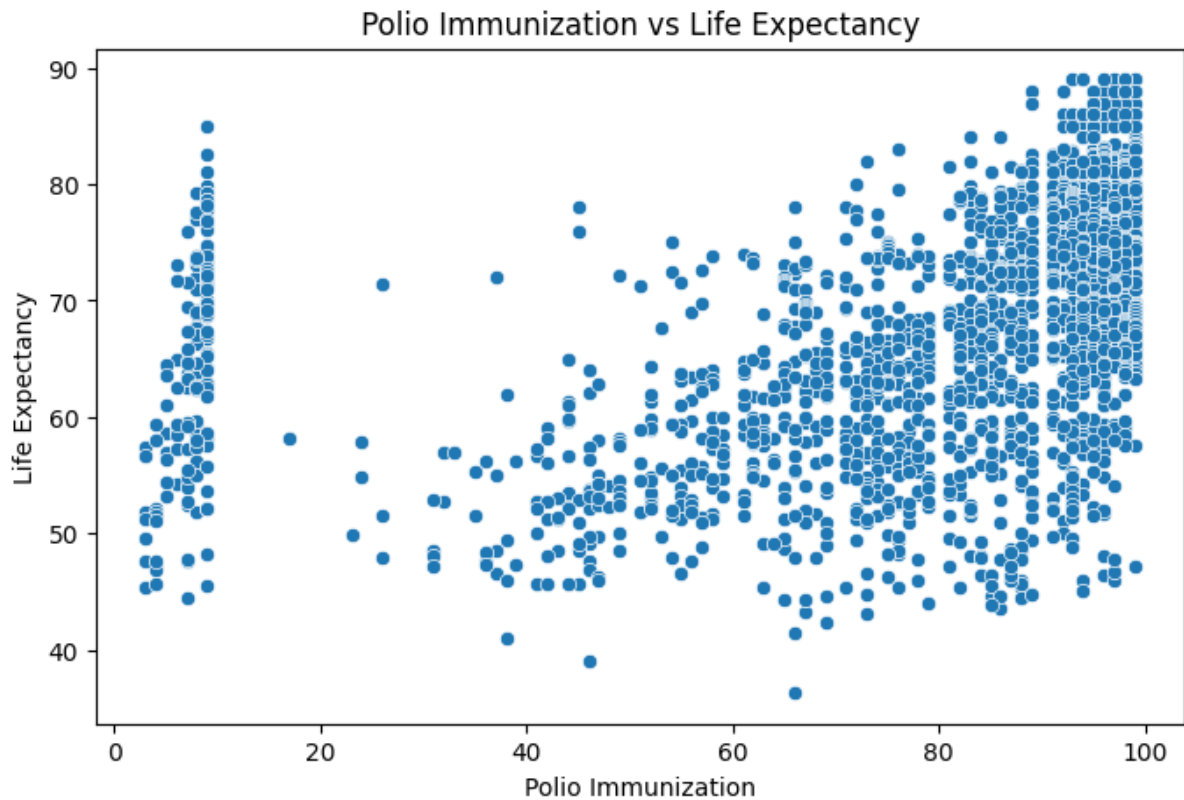










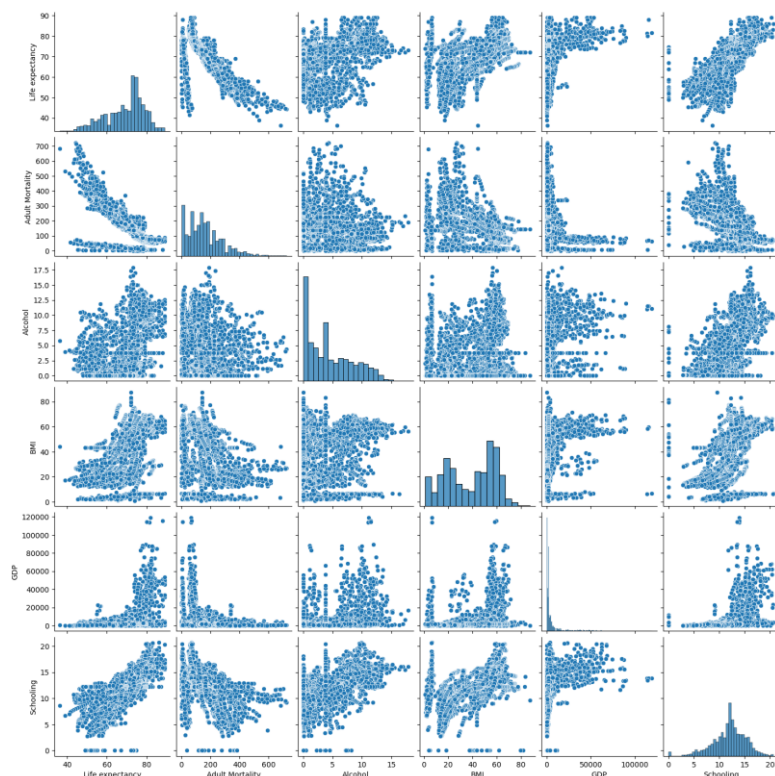
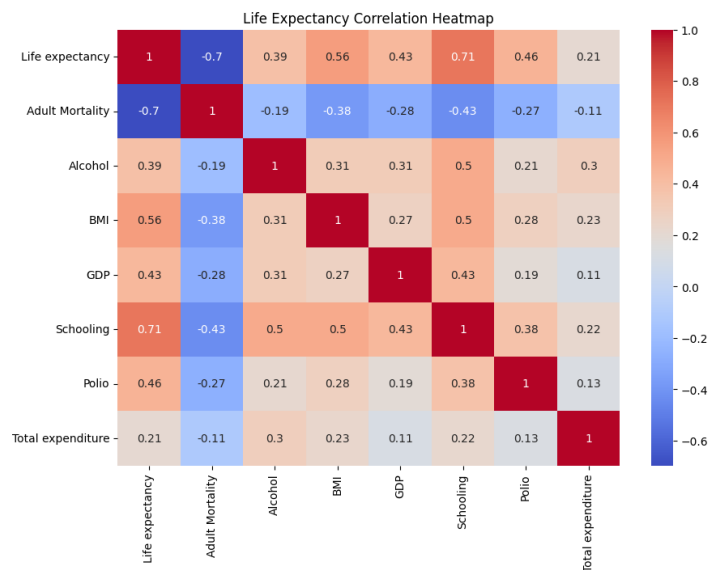


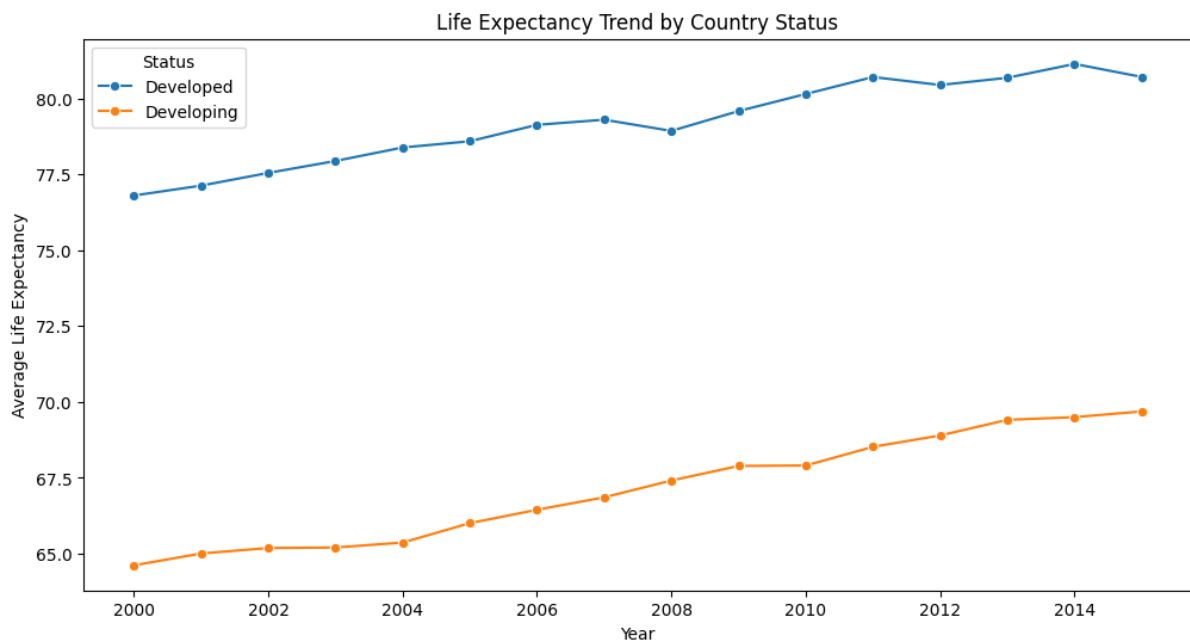
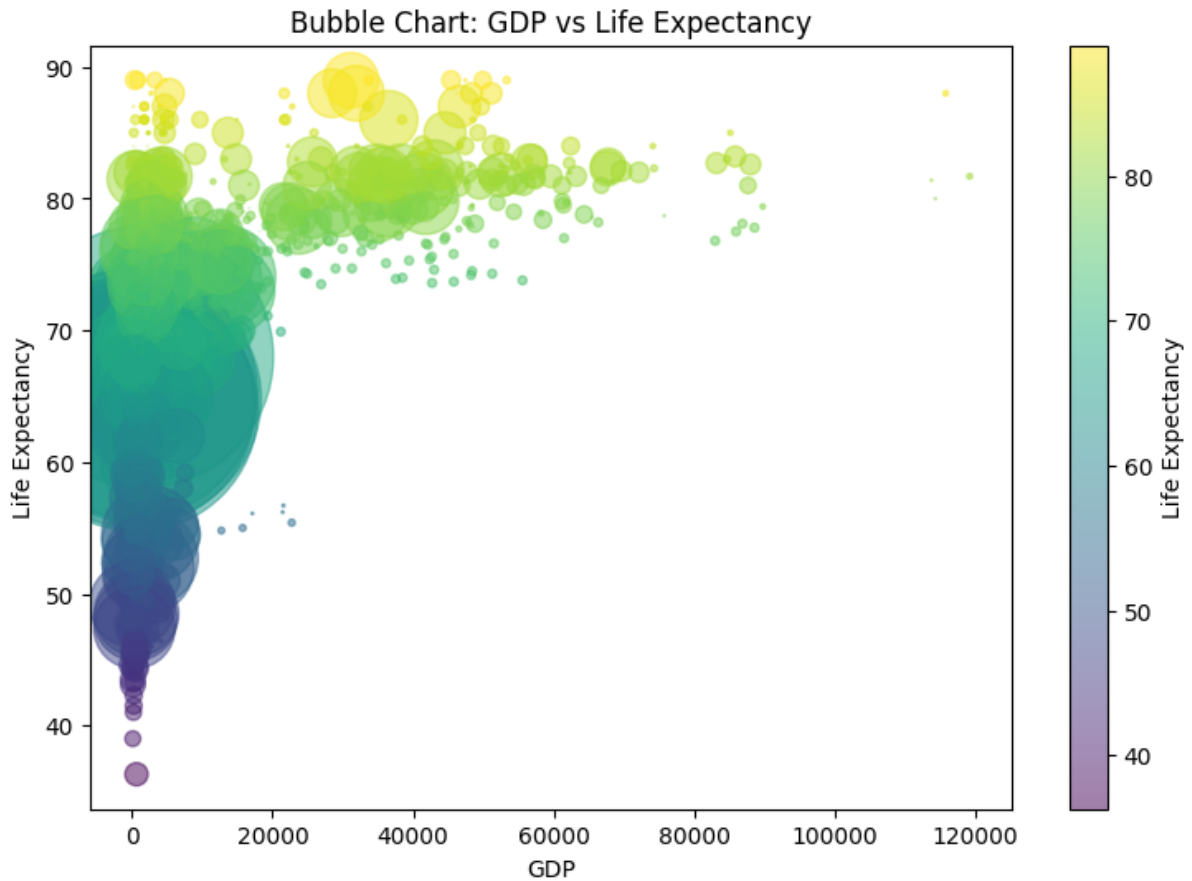
Correlation Matrix:

	Life expectancy	Adult Mortality	Alcohol	BMI \
Life expectancy	1.000000	-0.696390	0.388918	0.556901
Adult Mortality	-0.696390	1.000000	-0.190191	-0.379092
Alcohol	0.388918	-0.190191	1.000000	0.313022

BMI	0.556901	-0.379092	0.313022	1.000000
GDP	0.430461	-0.281434	0.313060	0.273932
Schooling	0.713054	-0.433415	0.498899	0.499806
Polio	0.458399	-0.270002	0.211744	0.281260
Total expenditure	0.208844	-0.111796	0.296153	0.230745

	GDP	Schooling	Polio	Total expenditure
Life expectancy	0.430461	0.713054	0.458399	0.208844
Adult Mortality	-0.281434	-0.433415	-0.270002	-0.111796
Alcohol	0.313060	0.498899	0.211744	0.296153
BMI	0.273932	0.499806	0.281260	0.230745
GDP	1.000000	0.432229	0.189926	0.112099
Schooling	0.432229	1.000000	0.375830	0.220170
Polio	0.189926	0.375830	1.000000	0.131037
Total expenditure	0.112099	0.220170	0.131037	1.000000





Top 10 Life Expectancy Records:

	Country	Year	Status	Life expectancy	GDP	Schooling
2056	Portugal	2014	Developed	89.0	2277.53613	16.8
995	Germany	2014	Developed	89.0	4792.65288	17.0
2433	Spain	2007	Developed	89.0	3279.41400	16.0
1850	New Zealand	2010	Developed	89.0	33691.34837	20.3
915	Finland	2014	Developing	89.0	49914.61864	17.0
241	Belgium	2014	Developed	89.0	47439.39684	16.3

937	France	2008	Developing	89.0	45413.65710	16.1
938	France	2007	Developing	89.0	416.58397	16.1
2513	Sweden	2007	Developed	89.0	53324.37937	15.8
1293	Italy	2004	Developed	89.0	31174.56118	15.6

Lowest 10 Life Expectancy Records:

	Country	Year	Status	Life expectancy	GDP	Schooling
1127	Haiti	2010	Developing	36.3	662.279518	8.6
2312	Sierra Leone	2000	Developing	39.0	139.314773	6.7
2311	Sierra Leone	2001	Developing	41.0	227.779471	7.0
2309	Sierra Leone	2003	Developing	41.5	263.761831	7.4
2308	Sierra Leone	2004	Developing	42.3	263.145817	7.6
1585	Malawi	2000	Developing	43.1	153.259487	10.7
2307	Sierra Leone	2005	Developing	43.3	287.689194	7.8
1584	Malawi	2001	Developing	43.5	146.761540	10.1
2921	Zambia	2000	Developing	43.8	341.955625	9.6
1583	Malawi	2002	Developing	44.0	29.979898	10.4

Encoded Status:

```
0 1
1 1
2 1
3 1
4 1
```

Name: Status, dtype: int64

Country Encoded Data:

	Afghanistan	Albania	Algeria	Angola	Antigua and Barbuda	Argentina \
0	True	False	False	False	False	False
1	True	False	False	False	False	False
2	True	False	False	False	False	False
3	True	False	False	False	False	False
4	True	False	False	False	False	False

	Armenia	Australia	Austria	Azerbaijan	...	United Republic of Tanzania \
0	False	False	False	False	...	False
1	False	False	False	False	...	False
2	False	False	False	False	...	False
3	False	False	False	False	...	False
4	False	False	False	False	...	False

	United States of America	Uruguay	Uzbekistan	Vanuatu \
0	False	False	False	False
1	False	False	False	False
2	False	False	False	False
3	False	False	False	False
4	False	False	False	False

	Venezuela (Bolivarian Republic of)	Viet Nam	Yemen	Zambia	Zimbabwe
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False

[5 rows x 193 columns]

Normalized Numeric Data:

	Country	Year	Status	Life expectancy	Adult Mortality \
0	Afghanistan	1.000000	1.0	0.544592	0.362881

1	Afghanistan	0.933333	1.0	0.447818	0.373961
2	Afghanistan	0.866667	1.0	0.447818	0.369806
3	Afghanistan	0.800000	1.0	0.440228	0.375346
4	Afghanistan	0.733333	1.0	0.434535	0.379501

	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles ... \
0	0.034444	0.0	0.003659	0.653061	0.005439 ...
1	0.035556	0.0	0.003774	0.622449	0.002319 ...
2	0.036667	0.0	0.003759	0.642857	0.002027 ...
3	0.038333	0.0	0.004014	0.673469	0.013135 ...
4	0.039444	0.0	0.000364	0.683673	0.014200 ...

	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population \
0	0.031250	0.452118	0.649485	0.0	0.004889	0.026074
1	0.572917	0.453279	0.618557	0.0	0.005127	0.000253
2	0.614583	0.450377	0.639175	0.0	0.005287	0.024525
3	0.666667	0.473012	0.670103	0.0	0.005608	0.002857
4	0.677083	0.435287	0.680412	0.0	0.000519	0.002302

	thinness 1-19 years	thinness 5-9 years	Income composition of resources \
0	0.619565	0.603509	0.505274
1	0.630435	0.610526	0.502110
2	0.637681	0.617544	0.495781
3	0.644928	0.628070	0.488397
4	0.655797	0.635088	0.478903

	Schooling
0	0.487923
1	0.483092
2	0.478261
3	0.473430
4	0.458937

[5 rows x 22 columns]

Cleaned dataset saved successfully.

Final Dataset Shape : (2938, 22)

Final Missing Values:

Country	0
Year	0
Status	0
Life expectancy	0
Adult Mortality	0
infant deaths	0
Alcohol	0
percentage expenditure	0
Hepatitis B	0
Measles	0
BMI	0
under-five deaths	0
Polio	0
Total expenditure	0
Diphtheria	0
HIV/AIDS	0
GDP	0
Population	0
thinness 1-19 years	0
thinness 5-9 years	0

Income composition of resources 0
Schooling 0
dtype: int64

Life Expectancy Data Analysis Completed Successfully.

RESULT